

# Cross-Modal Bayesian Reward Filtering for Robust Reinforcement Learning from Noisy Human Feedback

Anika Prakash\*

Rohan Pandey†

## Abstract

Reinforcement learning from human feedback (RLHF) gives agents access to dense supervisory signals when the environment reward is sparse, delayed, or difficult to specify. However, when human feedback is noisy, biased, fatigued, or adversarial, the agent may optimize the wrong signal confidently. This paper studies whether sparse environment rewards and simple environment-derived progress checks can be used to verify human feedback during training. We introduce the Cross-Modal Bayesian Reward Filter (BRF), an online reward-fusion mechanism that maintains a Beta posterior over the reliability of a human feedback channel. At each step, BRF compares the sign of human feedback with the sign of an environment-derived potential difference, updates a trust score, and uses that score to weight the human reward given to the agent. We evaluate BRF across 270 training runs spanning 3 MiniGrid environments, 3 reward modes (Sparse, Naive, and BRF), 6 synthetic teacher types, and 5 random seeds. On MiniGrid-DoorKey-8x8, where Sparse RL achieves 0.000 success universally, BRF reaches 0.460 under a Realistic teacher (+121% over Naive RLHF) and 0.404 under DistanceNoise (+84%). On Empty-8x8, BRF recovers to 0.600 under an Adversarial teacher while Naive RLHF fails at 0.000. Both human-feedback methods fail under extreme noise (80% flip probability), where only the Sparse baseline survives. LavaGap results also show that a potential function misaligned with the task’s safety constraints can corrupt the trust signal, allowing Sparse RL to dominate over human-feedback conditions. These results suggest that Bayesian arbitration can provide a low-cost safety layer for preference-based reinforcement learning when sparse but trustworthy environment information is available.

## 1 Introduction

Human feedback is often used when the reward is difficult to define. In reinforcement learning, an agent learns by maximizing cumulative reward while interacting with an environment [14]. In deep RLHF, a human supervisor can compare trajectory segments or provide scalar feedback, and a policy is then optimized against this learned or supplied reward signal [3, 15]. This approach has become popular because it replaces hand-designed reward functions with feedback that humans can more easily provide.

However, using human feedback introduces risk. While sparse environment rewards can be incomplete, they are usually tied to the task. Human feedback is denser and more informative, but can be noisy. A human teacher may become fatigued, misunderstand the objective, over-reward locally appealing behavior, or provide adversarial labels. These concerns are closely related to reward hacking, scalable supervision, and value-alignment failures studied in prior safety and preference-learning work [1, 9, 6]. If the agent treats every human label as ground truth, it has no mechanism to deal with inaccurate labels.

---

\*Student author. Email: [prakashanika5@gmail.com](mailto:prakashanika5@gmail.com).

†Corresponding author. Email: [rpande.1729@gmail.com](mailto:rpande.1729@gmail.com).

This paper investigates: Can an RL agent limit the effects of biased or noisy human feedback by using sparse environment rewards as a probabilistic reality check? We investigated this through the development of the Cross-Modal Bayesian Reward Filter. The filter treats the human as a noisy sensor, maintaining a posterior estimate of how often human feedback agrees with a separate environment-derived signal of progress. Frequent agreement means the human reward is trusted. When disagreement accumulates, the human reward is muted.

Our contribution is threefold. First, we formulate a bounded Bayesian trust update for online reward fusion between sparse environment rewards and dense human feedback. Second, we implement the filter as a wrapper that can be placed around standard reinforcement learning algorithms. Third, we evaluate the method in MiniGrid under several synthetic human-noise models, including stochastic, distance-dependent, fatigue, realistic, and adversarial feedback.

Results are environment-dependent. BRF shows substantial gains over Naive RLHF on tasks where the potential function is well-calibrated and teacher noise is moderate, but fails under extreme corruption and potential signals that conflict with the task’s safety constraints. The failure modes are as informative as the successes.

## 2 Related Work

**Reinforcement learning from human feedback.** Christiano et al. [3] showed that deep RL agents can learn complex behaviors from human preferences over trajectory segments while using only a small fraction of direct human feedback. Scalar human feedback has also been studied as a direct training signal, including in the TAMER and Deep TAMER line of work [15]. Later work has extended feedback-based reward learning to demonstrations, preferences, and richer annotation pipelines [10, 17]. These methods are powerful because human judgments provide dense task information. They are also vulnerable because the learned or supplied reward may be an imperfect proxy for the desired behavior.

**Reward-model errors and overoptimization.** When a policy is optimized against a learned reward model, errors in that model can be exploited. Previous safety work identifies reward hacking as a central accident risk [1], and more recent RLHF studies directly measure reward-model overoptimization [6]. Reward-model ensembles and conservative objectives can reduce these failures [4, 5]. BRF responds to the same concern, but acts at a different location in the pipeline. Rather than building multiple reward models, it arbitrates between a human channel and a sparse environment-derived channel online.

**Bayesian reinforcement learning.** Bayesian RL provides a principled way to represent uncertainty in models, values, or policies [8]. BRF uses a much smaller Bayesian object: a Beta posterior tracking the reliability of the human feedback. The purpose is not to explore MDP dynamics, but rather to determine how much to trust the supervisory signal.

**Reward shaping.** Potential-based reward shaping can accelerate learning while preserving optimal policies in suitable conditions [11, 16]. In our BRF, the change in potential at each step is used to measure the reliability of human feedback instead of as a standalone reward stream. The potential is based on progress towards the current goal in MiniGrid.

### 3 Method

The Bayesian Reward Filter (BRF) estimates teacher reliability online and uses it to weight the human reward channel, reducing the effect of biased feedback.

At each timestep  $t$ , the environment returns a sparse reward  $r_{\text{env},t} \in \{0, 1\}$  indicating whether the goal has been reached or not. While the environment’s reward is always accurate, its infrequency limits its usefulness. A human teacher simultaneously provides a scalar feedback signal  $r_{\text{human},t} \in [-0.1, 0.1]$ . Positive feedback indicates the transition moved the agent closer to the goal; negative feedback means the opposite.

#### 3.1 Cross-Modal Bayesian Filter

The reward delivered to the agent is

$$r_t = r_{\text{env},t} + w_t \cdot r_{\text{human},t}, \quad (1)$$

where  $w_t \in [0, 1]$  is a trust score which indicates the human’s reliability. The sparse reward remains unweighted because it is treated as the ground truth. The human feedback, which is dense but fallible, is weighted based on its accuracy historically.

The central design question is how to update  $w_t$ . At each step, the BRF compares the sign of the human feedback with that of the potential difference

$$\Delta\phi_t = \phi(s_{t+1}) - \phi(s_t). \quad (2)$$

$\phi$  is a potential function that quantifies progress for a given environment. Sign agreement means the human feedback and the environment-derived progress signal point in the same direction, and the trust weight is increased appropriately. Sign disagreement means the trust weight is decreased.

$$z_t = \begin{cases} 1, & \text{sign}(r_{\text{human},t}) = \text{sign}(\Delta\phi_t), \\ 0, & \text{sign}(r_{\text{human},t}) \neq \text{sign}(\Delta\phi_t). \end{cases} \quad (3)$$

No Bayesian update is applied when  $\Delta\phi_t = 0$ , as there is no new information about human reliability.

#### 3.2 Beta-Bernoulli Trust Update

BRF maintains a Beta posterior over teacher reliability, using the standard conjugate Beta-Bernoulli update for binary observations [7]:

$$\theta \sim \text{Beta}(\alpha_t, \beta_t).$$

The posterior parameters are initialized as  $\alpha_0 = \beta_0 = 1$ . This gives an initial trust weight  $w_t = \frac{1}{1+1} = 0.5$ , which means that there is no implicit assumption about human reliability; the human and the environment reward are given the same importance. The prior strength  $N_0 = \alpha_0 + \beta_0 = 2$  is intentionally weak, so the trust weight is quickly updated to reflect the current human’s reliability. For informative transitions, the update is

$$\alpha_{t+1} = \alpha_t + z_t, \quad \beta_{t+1} = \beta_t + (1 - z_t). \quad (4)$$

The trust score is the posterior mean:

$$w_t = \mathbb{E}[\theta \mid z_{1:t}] = \frac{\alpha_t}{\alpha_t + \beta_t}. \quad (5)$$

This update has two useful properties: First, the human reward is bounded rather than amplified. Second, trust changes online, so the same training run can respond to a teacher whose reliability changes over time.

### 3.3 Potential Functions in MiniGrid

For MiniGrid-Empty-8x8, the potential is

$$\Phi(s) = -d_{\text{Manhattan}}(p_{\text{agent}}, p_{\text{goal}}). \quad (6)$$

For LavaGap, the same distance term is augmented with a penalty near lava:

$$\Phi(s) = -d_{\text{Manhattan}}(p_{\text{agent}}, p_{\text{goal}}) - \lambda \cdot \text{lavaAdjacencyPenalty}. \quad (7)$$

For DoorKey, the potential is staged: the agent is first evaluated by distance to the key, then to the door, then to the goal.

$$\text{No key held: } \Phi(s) = -d_{\text{Manhattan}}(p_{\text{agent}}, p_{\text{key}}). \quad (8)$$

$$\text{Key held: } \Phi(s) = -d_{\text{Manhattan}}(p_{\text{agent}}, p_{\text{door}}). \quad (9)$$

$$\text{Door unlocked: } \Phi(s) = -d_{\text{Manhattan}}(p_{\text{agent}}, p_{\text{goal}}). \quad (10)$$

These potentials are not intended to be perfect task rewards. Instead, they are lightweight progress sensors that allow the filter to check if the human feedback is directionally plausible.

### 3.4 Baselines

We compare three reward-integration rules:

Table 1: Reward-integration methods.

| Method | Trust weight                            | Reward supplied to agent                          |
|--------|-----------------------------------------|---------------------------------------------------|
| Sparse | $w_t = 0$                               | $r_{\text{env},t}$                                |
| Naive  | $w_t = 1$                               | $r_{\text{env},t} + r_{\text{human},t}$           |
| BRF    | $w_t = \alpha_t / (\alpha_t + \beta_t)$ | $r_{\text{env},t} + w_t \cdot r_{\text{human},t}$ |

The Sparse baseline measures what can be learned without human feedback. The Naive baseline measures the standard failure mode in which the human channel is always trusted. BRF tests whether online trust estimation improves this tradeoff.

## 4 Experimental Setup

### 4.1 Environments

Experiments use MiniGrid [2], a family of grid-world environments used for goal-oriented RL. We evaluate on three tasks:

- **MiniGrid-Empty-8x8-v0**: The agent must reach a green goal tile in an empty room.
- **MiniGrid-LavaGapS7-v0**: The agent must reach the goal while avoiding a lava barrier.
- **MiniGrid-DoorKey-8x8-v0**: The agent must pick up a key, unlock a door, and reach the goal.

The tasks differ greatly in exploration difficulty. Empty-8x8 gives the cleanest test of noisy feedback because the shortest-path potential is informative. DoorKey is harder because the task requires staged behavior and has delayed consequences.

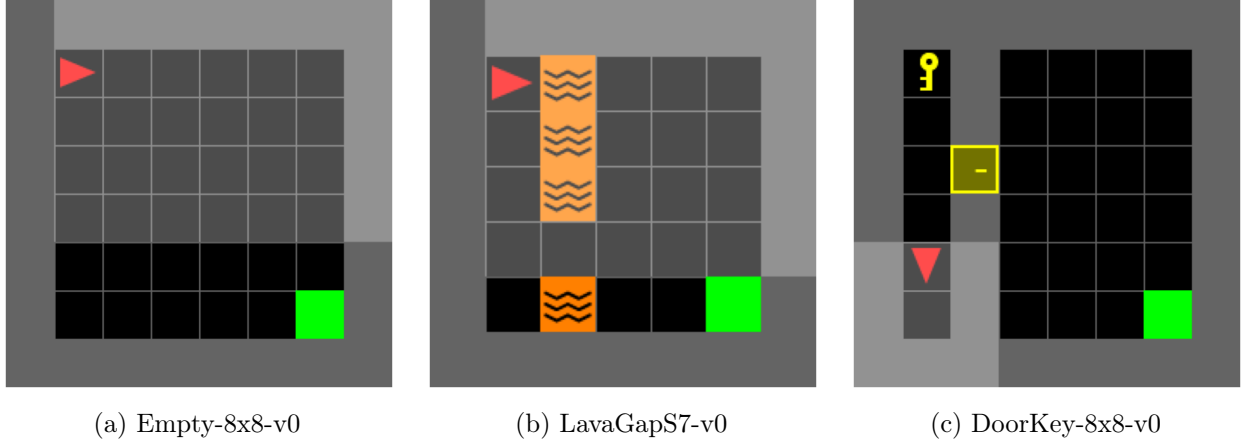


Figure 1: MiniGrid environment rendering [2].

## 4.2 Synthetic Human Teachers

We simulate several types of human teachers to represent different forms of unreliable feedback. All teachers return signals in  $[-0.1, 0.1]$ , keeping human feedback at 10% of the maximum terminal reward (+1.0). The goal is to separate recoverable noise from unrecoverable corruption.

- **StochasticLow:** Noise = 0.1. Feedback is flipped with a probability of 10%. Correct feedback 90% of the time.
- **StochasticHigh:** Noise = 0.8. Feedback is flipped with a probability of 80%.
- **Adversarial:** Feedback always contradicts the progress signal. Represents a worst-case attacker.
- **DistanceNoise:** Feedback becomes less reliable farther from the goal, with a maximum of 90% noise.
- **Fatigue:** Noise increases linearly from 0% to 80% over first 50,000 training steps, plateauing afterwards.
- **Realistic:** Combines distance confusion, temporal fatigue, and a base error rate of 5%.

## 4.3 RL Agents

All experiments use Proximal Policy Optimization (PPO) [13] as implemented in Stable-Baselines3 [12]. PPO is an on-policy actor-critic algorithm: at each iteration it collects a fresh batch of environment transitions, performs several gradient updates over that batch, then discards the data and repeats. On-policy agents are necessary for the BRF’s structure, since the BRF trust weight  $w_t$  is constantly updating as it learns about the teacher’s reliability. Because PPO trains exclusively on freshly collected transitions, every gradient update uses rollouts shaped only by the current trust estimate.

In contrast, off-policy algorithms like DQN store transitions in a replay buffer and sample from them uniformly throughout the training history. A transition collected before the trust estimate has converged carries a reward modified with an inaccurate  $w_t$ . That reward is fixed at storage time and cannot be corrected retroactively.

Therefore, the BRF’s per-step corrections are immediately effective for on-policy algorithms and only partially effective for off-policy ones. For this reason, only PPO was used for testing, and not agents such as DQN. BRF’s interaction with off-policy algorithms remains a direction for future work.

#### 4.4 Hyperparameters

All PPO hyperparameters use Stable-Baselines3 defaults (Table 2). No hyperparameter search was conducted on purpose. Tuning RL hyperparameters separately for each reward mode would introduce a confound, as any performance difference could be attributed to the tuning advantage rather than the reward structure.

SB3 defaults are well-established for MiniGrid-scale tasks and have been used without modification in prior MiniGrid benchmarking work, providing an additional basis for this choice.

Table 2: PPO hyperparameters (shared across all reward modes and all environments).

| Parameter                | Value                  | Role                                        |
|--------------------------|------------------------|---------------------------------------------|
| Total timesteps          | 100,000                | Training budget per run                     |
| Evaluation episodes      | 50                     | Post-training success rate estimate         |
| Random seeds             | 42, 123, 456, 789, 999 | 5 independent runs per condition            |
| Rollout length           | 2,048 steps            | Transitions collected per update cycle      |
| Epochs per rollout       | 10                     | Gradient passes over each batch             |
| Mini-batch size          | 64                     | Samples per gradient step                   |
| Learning rate            | $3 \times 10^{-4}$     | Adam optimizer step size                    |
| Discount factor $\gamma$ | 0.99                   | Weight on future vs. immediate reward       |
| GAE $\lambda$            | 0.95                   | Advantage estimator bias/variance trade-off |
| PPO clip range           | 0.2                    | Maximum policy update per step              |
| Value function coeff.    | 0.5                    | Critic loss weight in combined objective    |
| Entropy coefficient      | 0.0                    | No entropy bonus (task is well-defined)     |

**Network architecture.** Both the actor and critic use a shared `MlpPolicy` backbone: two fully-connected hidden layers of 64 units each with tanh activations, followed by separate output heads for the policy logits and the value estimate. The input is the flattened MiniGrid observation produced by `FlatObsWrapper`, which converts the image-based state tensor to a 1-D vector suitable for MLP processing. MiniGrid’s action space is `Discrete(7)`, which PPO can handle natively.

## 5 Results

The experiment matrix produced 270 training runs (1 PPO agent  $\times$  3 environments  $\times$  3 reward modes  $\times$  6 synthetic human teachers  $\times$  5 random seeds), each trained for 100,000 steps and evaluated over 50 episodes. Tables 3-5 report success rates averaged across seeds with the bold values showing the best-performing mode for each row.

### 5.1 Trust Weight Evolution

Figure 2 shows how the Bayesian trust weight  $w_t = \frac{\alpha_t}{\alpha_t + \beta_t}$  evolves over 100,000 training steps for each synthetic teacher. All curves begin at the uniform prior  $w_0 = 0.5$  and diverge within the first

few thousand steps.

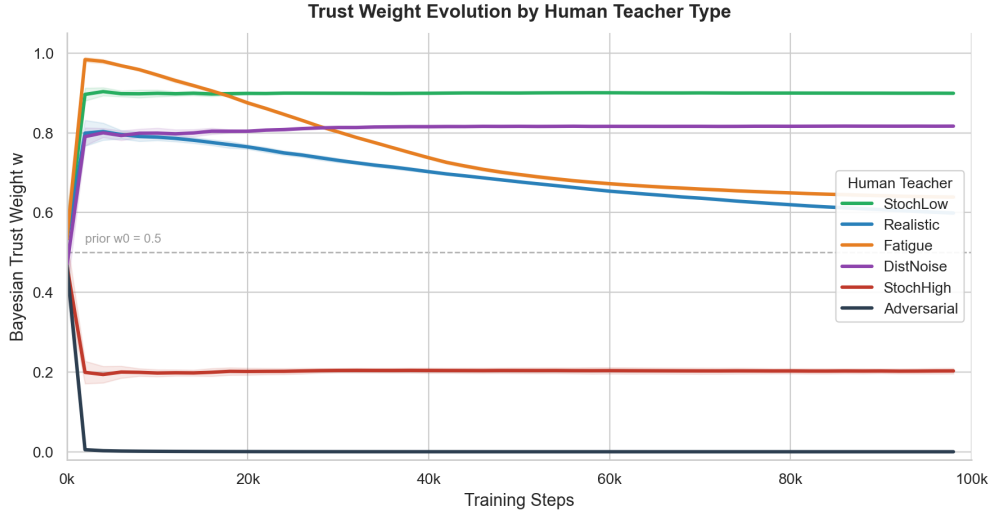


Figure 2: BRF’s trust weight evolution on MiniGrid-Empty-8x8-v0. Shows mean  $\pm 1$  SD across 5 seeds. Other environments show similar evolution trends.

The Adversarial teacher causes the trust score to collapse towards 0, as the feedback signal consistently opposes the potential function. The StochasticHigh teacher’s (80% noise) trust score stabilizes around 0.2, showing it provides accurate feedback 20% of the time. In contrast, the StochasticLow trust score stabilizes near a higher value of 0.9. The Realistic and Fatigue teachers become more inaccurate as training progresses, corresponding to the negative evolution slope. The DistanceNoise teacher becomes more accurate as the agent approaches the goal. The agent moves around during training instead of systematically converging, so the average distance to the goal is roughly constant and the trust curve is correspondingly flat.

## 5.2 MiniGrid-Empty-8x8-v0

Human feedback in the Naive and BRF modes allows the reinforcement learning model to achieve greater success under minimum noise. However, extreme corruption in the StochasticHigh and Adversarial teacher feedback reduces learning. BRF is able to recover slightly under Adversarial conditions, improving over the Naive baseline by 60 percentage points.

Table 3: PPO average success rates on MiniGrid-Empty-8x8-v0.

| Human          | Sparse       | Naive        | BRF          |
|----------------|--------------|--------------|--------------|
| StochasticLow  | 0.800        | <b>1.000</b> | <b>1.000</b> |
| Realistic      | 0.800        | <b>1.000</b> | <b>1.000</b> |
| Fatigue        | 0.800        | <b>1.000</b> | <b>1.000</b> |
| DistanceNoise  | 0.800        | <b>1.000</b> | <b>1.000</b> |
| StochasticHigh | <b>0.800</b> | 0.000        | 0.000        |
| Adversarial    | <b>0.800</b> | 0.000        | 0.600        |

### 5.3 MiniGrid-LavaGapS7-v0

The Sparse baseline unexpectedly outperformed human-feedback conditions in five of six LavaGap scenarios. The Sparse mode ignores human feedback completely, which is why it achieves a success rate of 0.244 across all teachers. The consistently lower performance of human-feedback methods reflects a potential function design issue involving the lava adjacency penalty. A teacher correct about goal-approach direction may simultaneously conflict with lava avoidance, making the cross-modal comparison in the trust update unreliable.

Once again, under Adversarial conditions, the BRF (0.160) recovers partially while Naive fails completely (0.000). This confirms the filter’s ability to suppress adversarial corruption.

Table 4: PPO average success rates on MiniGrid-LavaGapS7-v0.

| Human          | Sparse       | Naive        | BRF   |
|----------------|--------------|--------------|-------|
| StochasticLow  | 0.244        | <b>0.300</b> | 0.252 |
| Realistic      | <b>0.244</b> | 0.188        | 0.228 |
| Fatigue        | <b>0.244</b> | 0.240        | 0.112 |
| DistanceNoise  | <b>0.244</b> | 0.240        | 0.152 |
| StochasticHigh | <b>0.244</b> | 0.000        | 0.000 |
| Adversarial    | <b>0.244</b> | 0.000        | 0.160 |

### 5.4 MiniGrid-DoorKey-8x8-v0

DoorKey is the most consequential result. The Sparse baseline achieves 0.000 success across all teachers and seeds, showing that human feedback is necessary to solve the three-stage task.

BRF significantly outperforms the Naive baseline for the Realistic and DistanceNoise teachers, achieving 121% and 84% gains respectively. These teachers produce noise that varies with state, which the Beta posterior tracks correctly. However, Naive outperforms BRF for StochasticLow and Fatigue (discussed in Section 6).

Table 5: PPO average success rates on MiniGrid-DoorKey-8x8-v0.

| Human          | Sparse | Naive        | BRF          |
|----------------|--------|--------------|--------------|
| StochasticLow  | 0.000  | <b>0.272</b> | 0.188        |
| Realistic      | 0.000  | 0.208        | <b>0.460</b> |
| Fatigue        | 0.000  | <b>0.144</b> | 0.012        |
| DistanceNoise  | 0.000  | 0.220        | <b>0.404</b> |
| StochasticHigh | 0.000  | 0.000        | 0.000        |
| Adversarial    | 0.000  | 0.000        | 0.000        |



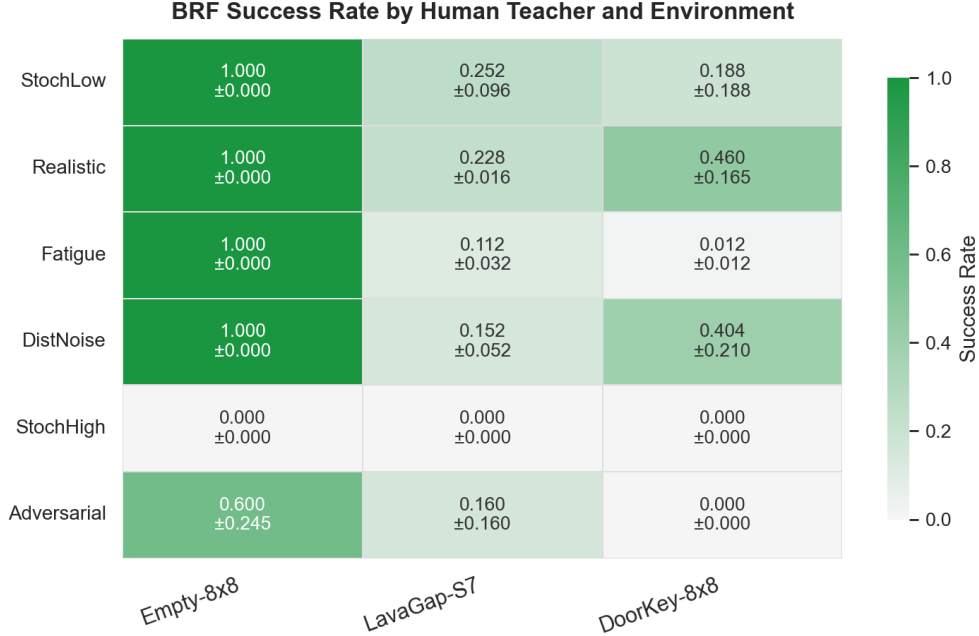


Figure 3: BRF success rate across all human teacher types and environments.

## 6 Analysis

**Human Feedback Necessity.** More complex tasks, such as DoorKey’s three-stage structure, proved to be unsolvable with sparse reward alone. DoorKey with sparse reward saw a 0.000 success rate over the 100,000 steps universally. Since feedback is necessary, its reliability must be gauged to determine whether feedback is helping or destroying learning. Naive RLHF ( $w = 1$ ) enables learning under StochasticLow (0.272) and DistanceNoise (0.220) but fails under Adversarial (0.000) and StochasticHigh (0.000) for DoorKey. BRF identifies teacher reliability without prior knowledge of which teacher is present.

**BRF Gains.** The BRF most significantly outperforms the Naive baseline on Empty-8x8 with the Adversarial teacher, with its success rate 0.600 higher. The filter drives trust to near-zero within the first few thousand steps, relying mainly on the sparse environment reward. The Sparse baseline still achieved a higher success rate of 0.800, because the BRF faced early damage before the trust fully collapsed. Additionally, BRF outperforms all baselines in DoorKey under the DistanceNoise and Realistic teachers. Both teachers produce noise that varies by state or accumulates from multiple sources, which the BRF tracks correctly with the Beta posterior. Notably, the BRF underperforms Naive on the StochasticLow teacher in DoorKey because BRF stabilizes around  $\approx 0.90$  and applies a persistent 10% handicap to the relatively reliable teacher.

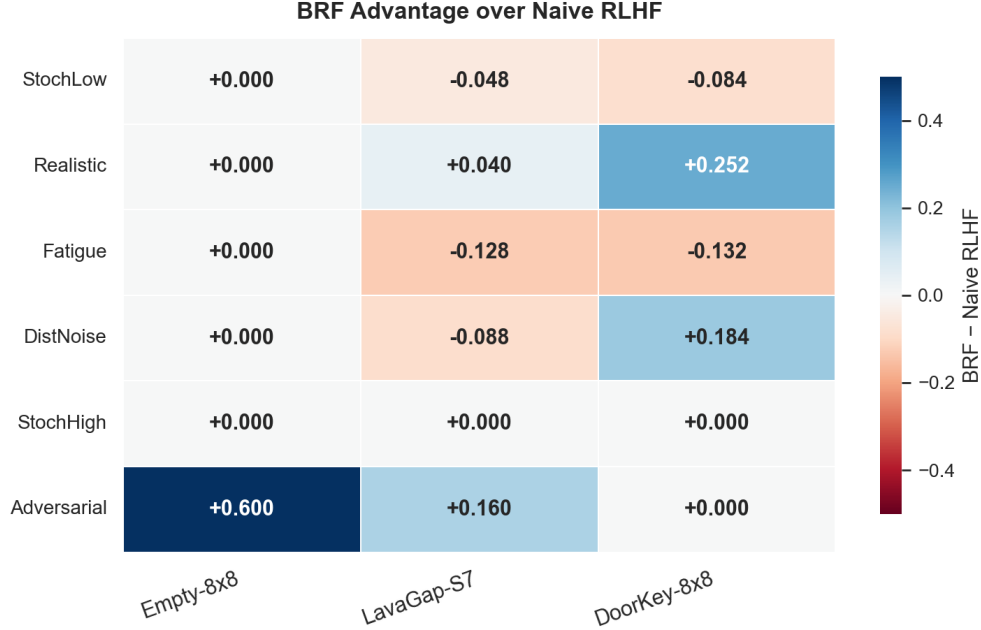


Figure 4: BRF success rate minus Naive RLHF success rate across all human–environment conditions. Blue means BRF wins; red means Naive wins.

**BRF Performance Anomalies.** BRF underperforms compared to Naive on DoorKey under Fatigue, achieving a success rate of 0.012 compared to Naive’s 0.144. Figure 2 shows BRF does detect the Fatigue teacher’s degrading reliability, but at a rate that is too slow. The large  $\alpha$  count accumulated during the reliable early phase (steps 0-25,000) slows the posterior’s response to rising noise. At step 50,000, when the teacher is already 80% noisy, trust is still  $\approx 0.65$ –0.70. This means the agent applies substantial weight to predominantly-wrong feedback at stages that are critical for DoorKey completion. Naive RLHF fully weights the human feedback throughout, which is more harmful than BRF in later noisy phases (Naive’s  $w = 1.0$  vs. BRF’s  $w \approx 0.65$ ). However, Naive still performs better, likely because Naive’s  $w = 1$  in the early reliable phase fully exploits correct feedback. This stronger policy foundation compounds in sequential stages, leading to better performance. BRF’s Beta-Bernoulli model can be slow to react to reliability changes. This limitation could be addressed with windowed or decay-weighted posteriors.

**Potential Function Sensitivity.** LavaGap shows BRF is limited by potential function quality. The lava-penalized potential  $\Phi(s) = -d_{\text{Manhattan}}(p_{\text{agent}}, p_{\text{goal}}) - \lambda \cdot \text{lavaAdjacencyPenalty}$  creates conflicting signals: a teacher correct about goal-approach direction may disagree with the lava-avoidance component. The BRF’s trust update is inaccurate when  $\Delta\Phi$  does not encode true progress, resulting in the Sparse baseline performing best for five of six teacher conditions.

**Recoverable Noise Threshold.** Results across environments suggest the BRF’s filtering capability has a noise threshold. StochasticLow (10% noise) enables learning in all environments; StochasticHigh (80%) causes complete failure for BRF and Naive, with only the Sparse baseline surviving in Empty (success rate of 0.800) and LavaGap (0.244). The threshold lies between these values. Characterizing it as a function of environment complexity, training budget, and filter design is a future goal.

## 7 Limitations

The first limitation is scale. MiniGrid is useful for controlled diagnosis, but it is simpler than language-model RLHF or real robotic feedback. The second limitation is the hand-designed potential. BRF assumes access to an environment-derived progress signal that is more reliable than the human label, but such a signal may be unavailable or itself biased for many tasks. The third limitation is the stationarity assumption. The Beta-Bernoulli model is slow to adapt to teachers with changing reliability, and this negatively impacted the DoorKey-Fatigue result. The fourth limitation is that BRF has only been evaluated on the on-policy PPO agent. Off-policy algorithms like DQN are structurally incompatible with the current BRF because reward-modified transitions stored in the replay buffer carry rewards computed under earlier, potentially inaccurate trust weights.

## 8 Future Work

**Learned Value Function.** The most direct extension removes the potential function dependency. PPO’s actor-critic architecture maintains a value function  $V(s)$  that estimates expected return from each state. At each step, the sign of  $V(s') - V(s)$  shows whether the transition represents progress, and this sign can be used as a learned proxy for  $\Delta\Phi$ . Replacing  $\text{sign}(\Delta\Phi)$  with  $\text{sign}(V(s') - V(s))$  in the BRF trust update allows the system to work without domain knowledge. However, this introduces a bootstrap dependency: the critic must learn accurate value estimates before it can reliably evaluate the human, but accurate critic learning benefits from reliable human feedback. When  $V$  is uninformed earlier in training, trust-estimation is near random (similar to the initial BRF prior). As the critic sharpens, trust signal and filtering improves. Future work will focus on characterizing this bootstrap phase in terms of duration, sensitivity to noise, etc.

**Off-Policy Compatibility.** Off-policy algorithms such as DQN are excluded from this study because transitions stored in a replay buffer carry rewards computed under early, potentially inaccurate trust weights. A solution could be storing the raw human reward  $r_{\text{human},t}$  and trust weight  $w_t$  at collection time, then recomputing the modified reward  $r_t = r_{\text{env},t} + w_{t,\text{current}} \cdot r_h$  when the transition is sampled. This allows the post-convergence trust estimate to retroactively correct early corrupted transitions.

**Adaptive Trust Dynamics.** The DoorKey-Fatigue anomaly (BRF=0.012, Naive=0.144) illustrates that fixed-prior Bayesian updating is not universally optimal. When a teacher’s reliability degrades monotonically, the large  $\alpha$  accumulated during the reliable phase slows the posterior’s adaptation to rising noise. This could be addressed through a mechanism where the influence of older observations decays when recent agreement variance is high, allowing the posterior to adapt more quickly to a changing teacher.

**Noise Threshold Characterization.** Results suggest a recoverable noise threshold between 10% and 80% flip probability. Characterizing this threshold as a function of environment complexity, training budget, and filter design could help predict when RLHF is expected to improve outcomes. This could be attempted through a systematic sweep across noise levels.

**Validation with Real Human Feedback.** All results use synthetic teacher models. Validation requires a study where human participants provide real-time feedback during agent training. This would help answer whether BRF’s advantages under synthetic conditions transfer to human-generated

feedback, whether trust-weight trajectories under real annotators match the patterns predicted by the synthetic models, and more.

## 9 Conclusion

The Cross-Modal Bayesian Reward Filter (BRF) maintains a Beta-Bernoulli posterior over teacher reliability, updated each step by comparing the sign of human feedback against an environment-derived potential difference. The resulting trust weight is used to scale the human reward channel, attempting to mitigate the effects of biased or inaccurate human feedback. Across 270 training runs (3 environments  $\times$  3 modes  $\times$  6 teachers  $\times$  5 seeds), results are environment-dependent but consistent in direction.

On DoorKey-8x8, Sparse RL has a success rate of 0.000 universally, showing human feedback is necessary for complex tasks. BRF achieves 0.460 under Realistic (+121% vs. Naive) and 0.404 under DistanceNoise (+84% vs. Naive), outperforming all baselines. On Empty-8x8, BRF identifies the Adversarial teacher’s unreliability and recovers to 0.600 while Naive fails at 0.000. In LavaGap, the human-feedback methods’ inability to outperform Sparse RL demonstrates the need for an accurate potential function rather than a failure of the BRF.

This method is computationally cheap, interpretable, and easy to attach to existing RL pipelines. However, the current approach is limited by the need for a hand-designed potential function as a ground-truth comparator, its application only to on-policy algorithms, and the stationarity assumption of the Beta-Bernoulli model (seen in the DoorKey-Fatigue result). BRF is therefore best viewed as a lightweight reliability layer for RLHF systems that have access to sparse but trustworthy environment evidence.

## References

- [1] Dario Amodei, Chris Olah, Jacob Steinhardt, Paul Christiano, John Schulman, and Dan Mané. Concrete problems in AI safety. *arXiv preprint arXiv:1606.06565*, 2016. <https://arxiv.org/abs/1606.06565>.
- [2] Maxime Chevalier-Boisvert, Bolun Dai, Mark Towers, Rodrigo de Lazcano, Lucas Willems, Salem Lahlou, Suman Pal, Pablo Samuel Castro, and Jordan Terry. MiniGrid and MiniWorld: Modular and customizable reinforcement learning environments for goal-oriented tasks. *arXiv preprint arXiv:2306.13831*, 2023. <https://arxiv.org/abs/2306.13831>.
- [3] Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences. In *Advances in Neural Information Processing Systems*, volume 30, 2017. <https://arxiv.org/abs/1706.03741>.
- [4] Thomas Coste, Usman Anwar, Robert Kirk, and David Krueger. Reward model ensembles help mitigate overoptimization. In *International Conference on Learning Representations*, 2024. <https://arxiv.org/abs/2310.02743>.
- [5] Jacob Eisenstein, Chirag Nagpal, Alekh Agarwal, Ahmad Beirami, Alex D’Amour, D. J. Dvijotham, Adam Fisch, Katherine Heller, Stephen Pfohl, Deepak Ramachandran, Peter Shaw, and Jonathan Berant. Helping or herding? reward model ensembles mitigate but do not eliminate reward hacking. *arXiv preprint arXiv:2312.09244*, 2023. <https://arxiv.org/abs/2312.09244>.

- [6] Leo Gao, John Schulman, and Jacob Hilton. Scaling laws for reward model overoptimization. In *Proceedings of the 40th International Conference on Machine Learning*, 2023. <https://arxiv.org/abs/2210.10760>.
- [7] Andrew Gelman, John B. Carlin, Hal S. Stern, David B. Dunson, Aki Vehtari, and Donald B. Rubin. *Bayesian Data Analysis*. CRC Press, 3 edition, 2013.
- [8] Mohammad Ghavamzadeh, Shie Mannor, Joelle Pineau, and Aviv Tamar. Bayesian reinforcement learning: A survey. *Foundations and Trends in Machine Learning*, 8(5-6):359–483, 2015. <https://arxiv.org/abs/1609.04436>.
- [9] Dylan Hadfield-Menell, Anca Dragan, Pieter Abbeel, and Stuart Russell. Cooperative inverse reinforcement learning. In *Advances in Neural Information Processing Systems*, volume 29, 2016. <https://arxiv.org/abs/1606.03137>.
- [10] Borja Ibarz, Jan Leike, Tobias Pohlen, Geoffrey Irving, Shane Legg, and Dario Amodei. Reward learning from human preferences and demonstrations in Atari. In *Advances in Neural Information Processing Systems*, volume 31, 2018. <https://arxiv.org/abs/1811.06521>.
- [11] Andrew Y. Ng, Daishi Harada, and Stuart Russell. Policy invariance under reward transformations: Theory and application to reward shaping. In *Proceedings of the 16th International Conference on Machine Learning*, pages 278–287, 1999.
- [12] Antonin Raffin, Ashley Hill, Adam Gleave, Anssi Kanervisto, Maximilian Ernestus, and Noah Dormann. Stable-Baselines3: Reliable reinforcement learning implementations. *Journal of Machine Learning Research*, 22(268):1–8, 2021. <https://jmlr.org/papers/v22/20-1364.html>.
- [13] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017. <https://arxiv.org/abs/1707.06347>.
- [14] Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction*. MIT Press, 2 edition, 2018.
- [15] Garrett Warnell, Nicholas Waytowich, Vernon Lawhern, and Peter Stone. Deep TAMER: Interactive agent shaping in high-dimensional state spaces. *Proceedings of the AAAI Conference on Artificial Intelligence*, 32(1), 2018. <https://arxiv.org/abs/1709.10163>.
- [16] Eric Wiewiora. Potential-based shaping and Q-value initialization are equivalent. *Journal of Artificial Intelligence Research*, 19:205–208, 2003. <https://arxiv.org/abs/1106.5267>.
- [17] Yifu Yuan, Jianye Hao, Yi Ma, Zibin Dong, Hebin Liang, Jinyi Liu, Zhixin Feng, Kai Zhao, and Yan Zheng. Uni-RLHF: Universal platform and benchmark suite for reinforcement learning with diverse human feedback. In *International Conference on Learning Representations*, 2024. <https://arxiv.org/abs/2402.02423>.